

Aegis Home System — Draft v4 V2.0

Adding Building Envelope Shielding

Version 2.0 — June 2026

Iteration 4 extends protection to the building envelope — walls and windows.

Δ Change Log

- AgNW/PI transparent EMI film: 87% transmittance + 40 dB SE (2025)
- MXene alternative: $\text{Ti}_3\text{C}_2\text{T}_x$ achieving 55 dB SE with better chemical resistance
- Basalt-SiC wall blanket: updated composite density 2025

New V4 Components

10. **Basalt-SiC Wall Blanket (CSMFAB000000000011)** — RF absorption without DC conductivity; SiC fibers in basalt/aerogel matrix; integrates between drywall layers; SE = 45 dB at 1 GHz
11. **ITO-Free Window Shielding Film (CSMFAB000000000011)** — AgNW/PI transparent conductor; 87% visible transmittance + 40 dB SE; YInMn Blue laminate for NIR; installed over existing glass

Full building envelope coverage: - Walls: 45 dB SE (Basalt-SiC blanket) - Windows: 40 dB SE (transparent film) - Roof: YInMn Blue ceramic tiles (SRI=115) - Foundation: BFRP rebar, no GIC loop

V4 cumulative GIC threat reduction: ~82% (complete envelope)